

Bachelor's Programme in Science and Technology

Robust Fine-Grained Gesture Recognition via Continual Learning

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Abstract

Recently, radio frequency (RF)-based gesture recognition has gained popularity thanks to its applications. In stable and controlled environments with fixed settings (e.g., a fixed set of participants, gestures, and standing positions), RF-based gesture recognition systems typically face little challenges due to powerful sensors and accurate machine learning algorithms. However, in dynamic environments, such systems struggle to learn as the settings change continually (e.g., new participants and new gestures). Therefore, it is common for RF-based gesture recognition systems to adapt to new settings at the expense of past knowledge, a problem called catastrophic forgetting. Thus, continual learning emerges as a solution to the problem. Continual learning refers to the application of appropriate techniques to mitigate catastrophic forgetting, which helps the machine learning model to learn in dynamic environments. While there are multiple continual learning methods, this thesis evaluates the performance of Elastic Weight Consolidation (EWC), Synaptic Intelligence (SI), Learning without Forgetting (LwF), Experience Replay (ER), and Averaged Gradient Episodic Memory (A-GEM) when applied to an RF-based gesture recognition deep learning network. Although all methods are effective in addressing catastrophic forgetting, the results indicate that the performance of replay-based methods (i.e., ER and A-GEM) is better than the performance of regularization-based methods (i.e., EWC, SI, and LwF).

Keywords Catastrophic forgetting, continual learning, gesture recognition

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1 Introduction

A gesture recognition system is a system that utilizes sensors to capture motion data and identifies the gesture performed by the user, usually with a machine learning algorithm [36]. Due to recent developments in sensor technologies and machine learning algorithms, gesture recognition has become increasingly popular. Among the different types of gesture recognition methods, radio frequency (RF)-based methods stand out, as they typically do not involve wearable devices and do not capture users' visual data [32]. RF-based methods usually use radars to emit radio waves and receive their reflections to capture motion data [32].

One of the challenges for gesture recognition systems in particular and RF-based gesture recognition systems in general is the continual appearance of new data. The appearance of new data often arises from the introduction of new users, of new gestures, or of new environment settings. The incremental arrival of new data is poorly handled by traditional machine learning methods [28]. These methods either retrain the model with both old and new data, which is cost and time inefficient, or naive fine-tuning (i.e., continue training the model using only new data without any techniques to preserve past knowledge), which leads to catastrophic forgetting (see Subsection 2.1 for definition) [28], [41].

Continual learning, commonly referred to as incremental learning, is a promising solution to the challenge. Using continual learning strategies, gesture recognition systems can learn new training data without forgetting past knowledge [28]. Compared to retraining, continual learning enhances the time and cost efficiency of gesture recognition systems [41]. Compared to naive fine-tuning, continual learning improves accuracy and mitigates catastrophic forgetting [28]. Hence, continual learning is crucial for gesture recognition models, especially in dynamic environments.

Recent studies have explored the application of continual learning in gesture recognition [23], [30], [38], [39]. However, most of the research focuses on vision-based and inertial-based gesture recognition [23], [39]; this leaves continual learning in depth-based gesture recognition relatively unexplored. Therefore, this thesis aims to evaluate the performance of different continual learning methods in RF-based gesture recognition. The compared methods include two parameter regularization methods, a functional regularization method, and two replay-based methods. From the results of the experiments, the strengths and limitations of the compared methods are evaluated.

The remainder of the thesis is as follows. Section 2 provides an overview of continual learning and continual learning in gesture recognition. Section 3 outlines the compared methods and the structure of the dataset. Section 4 presents and analyzes the results of the experiments. Section 5 discusses the results. Section 6 concludes the thesis.

2 Literature Review

This section reviews existing work on continual learning and continual learning in gesture recognition. The section is divided into two subsections. Subsection 2.1

presents an overview of continual learning, introduces three types of continual learning, and discusses the main categories of continual learning methods. Subsection 2.2 provides an overview of gesture recognition and reviews existing studies related to continual learning in gesture recognition.

2.1 Continual Learning

2.1.1 Overview of Continual Learning

In traditional machine learning settings, it is typical that the entire training dataset is available from the beginning. In the training process, the task of a neural network is to capture the distribution of the training dataset. The captured data distribution is then used by the network to perform prediction on unseen data during inference. However, in **continual learning** settings, not all the training data is available from the beginning [41]. Instead, the data is provided by a non-stationary data stream (i.e., the data distribution varies through time) [29], [41], [42]. Hence, in such settings, it is more difficult for the network to capture the distribution of the training data when compared with traditional settings.

In continual learning settings, the ability to incrementally adapt to new training data while retaining learned information is crucial for a neural network [28]. Hence, continual learning methods often apply techniques to incorporate new training data while preserving past knowledge. Without these continual learning techniques, there are two possible alternatives. One of them is to retrain the model using both old and new data—a process that significantly increases the cost of an already expensive training process [41]. Furthermore, as retraining involves keeping prior training data, this method is difficult to implement in certain scenarios. For example, in health care, the data storage raises privacy concerns, which eliminate the possibility of retraining [35]. Another alternative is to naively continue the training of the model using the new data [28]. Since each batch of new data has a different distribution [42] (due to the non-stationary data stream), the naive approach would fail to capture the overall distribution; hence, it would face a rapid loss of previously learned knowledge [28]—a problem known as catastrophic forgetting [1]. These challenges promote the need for continual learning methods that can efficiently train models without having to depend on full training data storage or costly retraining.

There are two main focuses for a continual learning method: learning new data and preserving past knowledge [28]. The ability to learn new data is referred to as plasticity, and the ability to preserve past knowledge is referred to as stability [4]. For a neural network to excel in continual learning, it must find the balance point in the plasticity–stability dilemma [4]. A continual learning model not only needs stability to prevent catastrophic forgetting, but also needs plasticity to serve its purpose, which is learning [4]. However, since this is a trade-off, high stability means low plasticity and vice versa [34]. Hence, a model with high plasticity and low stability can quickly adapt to new data, but at the expense of past knowledge [4]. On the other hand, a model with low plasticity and high stability has difficulties in learning new data but can better retain past knowledge [4].

2.1.2 Three Types of Continual Learning

There are three types of continual learning: task-continual learning, domain-continual learning, and class-continual learning [29] (see Table 1 for an overview).

Scenario	Intuitive description	Mapping to learn
Task-incremental learning	Sequentially learn to solve a number of distinct tasks	$f : \mathcal{X} \times \mathcal{C} \rightarrow \mathcal{Y}$
Domain-incremental learning	Learn to solve the same problem in different contexts	$f : \mathcal{X} \rightarrow \mathcal{Y}$
Class-incremental learning	Discriminate between incrementally observed classes	$f : \mathcal{X} \rightarrow \mathcal{C} \times \mathcal{Y}$

Table 1: Three common incremental learning (i.e., continual learning) scenarios and their corresponding function mappings [29].

Task-continual learning is the form of continual learning where the system sequentially learns a set of tasks from a non-stationary data stream. The key property of task-continual learning is that the task attributes are provided to the model as input during training, testing, and inference [29]. This allows the implementation of task-specific components (e.g., output heads and encoders), or even the allocation of a separate network for each task [29]. In the latter case, catastrophic forgetting is effectively avoided, as there are no shared parameters between tasks [29]. Hence, the main goals of task-continual learning are to efficiently share parameters, to promote forward transfer (i.e., learning a new task using knowledge from previous ones), and to promote backward transfer (i.e., reinforcing earlier tasks by knowledge from the new task) [29], [44].

The next form of continual learning is **domain-continual learning**. In this form, the overall objective of the model remains constant, but the domain incrementally changes [29]. For instance, consider a domain-continual learning model with the overall objective of recognizing digits from 0 to 9. While the output space is fixed, each context provides inputs from a different domain—for example, handwritten digits, printed digits from scanned documents, or digits from digital clocks. The difference between domain-continual learning and task-continual learning is that, while in task-continual learning, the task-relevant information is provided during both training and inference, in domain-continual learning, the information about the input distribution (i.e., the domain) is typically not explicitly available at either stage [29].

The last form of continual learning is **class-continual learning**. As the name suggests, in class-continual learning, the algorithm learns to distinguish among a sequentially growing set of tasks [29]. In each context, the algorithm is only exposed to the training data of a number of classes [29]. An effective approach is to create a "prototype" (i.e., a latent representation) in the latent space for each class and utilize the distance between the latent representation of the input and the prototypes for prediction [12] (template-based methods) [41]. The main challenge of class-continual learning is that the algorithm needs to classify over the entire set of classes, unlike in task-continual learning, where the algorithm only needs to make predictions over

a subset of classes specified by the task. This explains the notable performance of replay-based methods and template-based methods in [29], as it would be problematic to distinguish a class in the current context from classes in past contexts without their examples or representations [29].

To formalize, according to the formulation in [29], let $\mathcal{X}$ be the input space, $\mathcal{Y}$ be the within-context output space, and $\mathcal{C}$ be the context space. The global output space can be determined with the within-context output space $\mathcal{Y}$ and the context space $\mathcal{C}$; therefore, let $\mathcal{G} = \mathcal{Y} \times \mathcal{C}$ be the global output space [29]. The mapping of three types of continual learning can be formalized as follows. The goal of task-continual learning is to learn the mapping $f : \mathcal{X} \times \mathcal{C} \rightarrow \mathcal{Y}$, as in each task, the input comes with its task-specific context, and the output space of the task is determined by the scope of that task [29]. Domain-continual learning aims to learn the mapping $f : \mathcal{X} \rightarrow \mathcal{Y}$ because the algorithm needs to capture the input distribution across domains, and the output space is fixed [29]. Class-continual learning follows the mapping $f : \mathcal{X} \rightarrow \mathcal{G}$, since it must perform the global classifications over the set of previously seen classes [29].

2.1.3 Continual Learning Methods Categories

Van de Ven et al. [41] classifies continual learning methods into 5 categories: context-specific components methods, template-based methods, parameter regularization methods, functional regularization methods, and replay-based methods. These five categories are briefly introduced below; however, the compared methods in this thesis fall only into the latter three categories.

Context-specific components methods allocate a portion of components to each task, which can be done by using the gating mechanism [18] or assigning to each task an entire separate network [41]. In the latter case, catastrophic forgetting is completely avoided; however, this either decreases the learning capacity of the model or increases the total number of model parameters [41].

In **template-based methods**, each class has a latent representation in the embedding space, which is obtained by using class-specific variational autoencoders (VAE) [27] or by taking the average of the embedding space representation of the stored examples [12].

In **parameter regularization methods**, a regularization term is included in the total loss [29]. This term discourages radical changes in important parameters, which effectively reduce the effect of catastrophic forgetting [41].

Similarly, the loss function of **functional regularization methods** also includes a regularization term [29]. This helps penalize changes in the model output from a set of anchor points as the input [29]. Anchor points can be obtained by using input from the current context [10] or retaining a portion of input from previous contexts [26].

Replay-based methods typically include retaining a small portion of the data from each class for replaying (i.e., rehearsal) [19], [21], [28] and capturing the latent distribution of each class to generate pseudo-datapoints for replaying (i.e., pseudo-rehearsal) [13], [22], [28].

2.2 Gesture Recognition and Continual Learning in Gesture Recognition

2.2.1 Overview of Gesture Recognition

Gesture recognition is the technology that tracks the movement of the human body, then identifies the gesture based on the set of registered gestures [37]. Gesture recognition systems usually utilize sensors and cameras to capture the body movements (typically the movement of arms, hands, and fingers) [46]. A recognition algorithm analyzes and attempts to recognize the gesture; then it translates the identified gesture to the corresponding command [46].

Gesture recognition has various significant applications. In the field of smart home [5], [6], [31], gesture recognition enables people to control their home applications (e.g., TVs, fans, and lights) using only the motion of their hands. In the field of gaming and entertainment, gesture-based controllers enable players to utilize their body movements to control the in-game characters, which improves the gaming experience. Furthermore, regarding sign language interpretation, gesture recognition-based systems allow individuals with hearing or speech impairments to communicate more smoothly and efficiently [33].

The approaches to collect gesture data can mainly be divided into three categories: inertia-based, vision-based, and depth-based. Some hybrid approaches use more than one method to improve accuracy and robustness [8]. **Inertia-based methods** typically involve users wearing or holding gadgets equipped with accelerometers and gyroscopes [17], [38]—such as hand gloves, smartphones, or motion controllers (e.g., virtual reality controllers, Wii Remote [47]). These gadgets effectively capture the users' hand movements in three-dimensional space without the impact of lighting or the environment [17]. Despite these advantages, since these methods require users to wear or hold devices, they lack convenience compared to other hands-free alternatives [3]. **Vision-based methods** utilize cameras to capture the user movements [3]. Even though cameras capture only the two-dimensional information, this is sufficient to recognize gestures in many cases. Nonetheless, some state-of-the-art pose-detection models (e.g., MediaPipe Hands [24] can efficiently extract depth-relevant information, enabling accurate gesture recognition). However, vision-based approaches are usually sensitive to background and lighting [3], making them unsuitable for certain environments. **Depth-based methods** leverage depth measurements relative to the signal source for gesture recognition, using wave sensors (e.g., mmWave sensor) for data collection [3], [32]. Similar to inertia-based methods, depth-based methods are not affected by lighting. However, their performance might degrade in dynamic environments with background motions. Additionally, one study [43] also considered biosignal-based methods [2], [25]—which analyze muscle activities to identify gestures—as a category.

2.2.2 Continual Learning in Gesture Recognition

The ability to learn from a non-stationary stream of data is critical for gesture recognition models, due to the need to add extra classes or to update the environments

of existing classes (e.g., indoor and outdoor). Hence, many recent studies focus on continual learning in gesture recognition.

In 2020, Wang et al. [23] presented ClAss incremenTal NETworks (CatNet). CatNet is the incorporation of 3D Convolutional Neural Network (3D ConvNet) and Incremental Classifier and Representation Learning (iCaRL) [12], [23]. The framework iCaRL utilizes a feature extractor to project the inputs to the embedding space [23]. During training, a portion of the training data (i.e., examples) from previous contexts is stored, which is used to calculate the "template" of each class by averaging these examples' embedding space representations [23]. During inference, the classification is based on the distance between the input and the class templates in the embedding space [23]. Moreover, the stored data is used in combination with the new data to train the feature extractor [23].

In 2023, Aich et al. [30] introduced a data-free (i.e., not retaining real data from past contexts) class-incremental hand gesture recognition algorithm. The method captures the mean and variance of previous classes in the embedding space, which are then used to generate prototypical support vectors (i.e., proto-SVs) [30]. A set of support vectors (SVs) is also identified using a support vector machine (SVM) [30]. Model inversion is then used to generate support inputs (SIs) from the combination of SVs and proto-SVs [30]. Finally, the model is fine-tuned using new class data and the SIs as the data of old classes [30].

In 2024, Park et al. [39] proposed the Continual Gesture Recognition System (CGRS), which is implemented using resources from a Samsung Galaxy S10 with limited memory. It aims to solve class-continual learning [39]. The framework effectively utilizes gyroscopes and accelerometers to identify the user's gesture [39]. CGRS leverages a size-limited buffer to store data from past contexts, which is then replayed in the current context to prevent catastrophic forgetting [39].

In 2024, Liu et al. [38] put forward iKAN, which is the first continual learning framework based on Kolmogorov-Arnold Network (KAN) [45]. It aims to solve task-continual learning [38]. The heart of iKAN is the classifier, which is designed based on KAN [38]. The classifier uses a spline-based architecture, allowing it to update a part of the network without affecting the globally learned structure, thanks to the local plasticity of splines [38]. This enables the system to adapt to new training data while avoiding catastrophic forgetting [38].

Overall, these papers highlighted the popularity of applying continual learning to gesture recognition systems.

3 Methodology

This section covers the methodology used in the experiments. The section is divided into two subsections. Subsection 3.1 demonstrates the architecture of the base network, to which continual learning methods are applied. Subsection 3.2 presents the formulas and algorithm details of the compared methods.

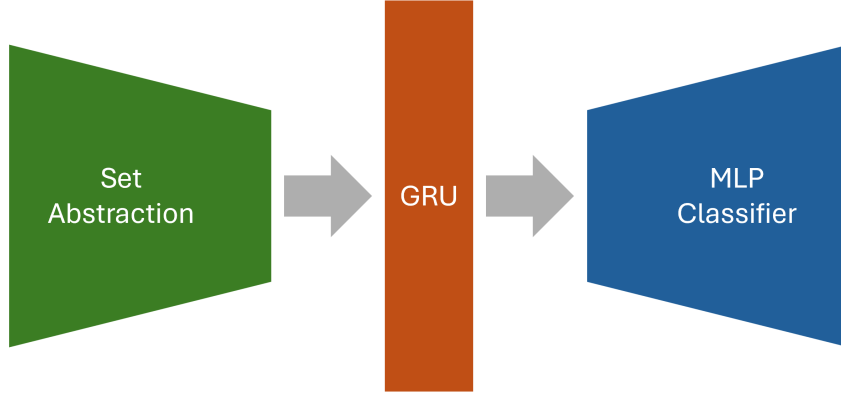


Figure 1: Base model architecture.

3.1 Base Model Architecture

There are three main components in the base network—the set abstraction module, the GRU layer, and the classifier (see Figure 1). The set abstraction module is adapted from the set abstraction layers of PointNet++ [11]. It takes a sequence of frames as input and applies sampling, grouping, and feature extraction repeatedly [11]. This results in a more compact sequence of frames, but it still contains important features of the original sequence. The GRU layer then processes the new sequence and outputs a vector that captures the geometric and temporal features across all frames. This vector is passed to the classifier to predict the class to which the original sequence belongs.

3.2 Continual Learning Methods

Baseline methods include None [41], Joint [41], elastic weight consolidation (EWC) [9], synaptic intelligence (SI) [14], learning without forgetting (LwF) [10], experience replay (ER) [20], [21], and averaged gradient episodic memory (A-GEM) [19].

None

None, which serves as the lower baseline, is the naive approach where the model incrementally learns the training data of each context [29]. In the None method, there is no attempt to mitigate catastrophic forgetting.

Joint

Joint is the higher baseline, where all the training data are available at the beginning [29]. This is similar to the traditional deep learning settings.

Elastic Weight Consolidation

For EWC, a regularization term $\mathcal{L}_{\text{EWC}}(\theta)$ is added to the total loss [9]. During training, in context K , $\mathcal{L}_{\text{EWC}}(\theta)$ can be formulated as:

$$\mathcal{L}_{\text{EWC}}(\theta) = \lambda \sum_{k=1}^{K-1} \left(\frac{1}{2} \sum_{i=1}^{N_{\text{params}}} F_i^{(k)} (\theta_i - \hat{\theta}_i^{(k)})^2 \right),$$

where λ is the regularization strength (set to 10 based on hyperparameter tuning; see Subsection 4.4 for details), $\hat{\theta}_i^{(k)}$ is the i -th parameter after the training of context k , θ_i is the current value of the i -th parameter, and $F_i^{(k)}$ is the Fisher information for parameter θ_i [9], [29]. The Fisher information of the i -th parameter measures the importance of the i -th parameter to the input-output mapping of the model in context k :

$$F_i^{(k)} = \frac{1}{|S^{(k)}|} \sum_{x \in S^{(k)}} F_{i,x}^{(k)},$$

with

$$F_{i,x}^{(k)} = \sum_o p_{\hat{\theta}^{(k)}}(o | x) \left(\frac{\delta}{\delta \theta_i} \log p_{\theta}(o | x) \Big|_{\theta=\hat{\theta}^{(k)}} \right)^2,$$

where $S^{(k)}$ is the input set in context k and $p_{\theta}(o | x)$ is the probability—predicted by the model with parameter θ —that input x belongs to class o [9], [29]. Thus, EWC effectively discourages significant changes in important parameters by minimizing the difference between the model parameters and context-optimized parameters $\hat{\theta}_i^{(k)}$ of each context k [9].

Synaptic Intelligence

In SI, the loss function includes a surrogate loss $\mathcal{L}_{\text{SI}}(\theta)$ [14]. The surrogate loss during context $K > 1$ is defined as

$$\mathcal{L}_{\text{SI}}(\theta) = \lambda \sum_i^{N_{\text{params}}} \Omega_i^{(K-1)} (\tilde{\theta}_i - \theta_i)^2,$$

where λ is the regularization strength (set to 0.1 based on hyperparameter tuning; see Subsection 4.4 for details), $\tilde{\theta}_i$ is the i -th parameter at the beginning of the current context, θ_i is the current value of the i -th parameter, and $\Omega_i^{(K-1)}$ is the importance of the i -th parameter [14], [29]. The importance $\Omega_i^{(K-1)}$ is updated after the training of each context as follows:

$$\Omega_i^{(K-1)} = \sum_{k=1}^{K-1} \frac{\omega_i^{(k)}}{\left(\hat{\theta}_i^{(k)} - \hat{\theta}_i^{(k-1)} \right)^2 + \xi},$$

where $\hat{\theta}_i^{(k)}$ is the i -th parameter after the training of context k , ξ is set to 0.1 to avoid zero division, and $\omega_i^{(k)}$ is updated after each iteration according to the formula

$$\omega_i^{(k)} = \sum_{t=1}^{N_{\text{iters}}} \left(\theta_i^{(k)}[t] - \theta_i^{(k)}[(t-1)] \right) \frac{-\delta}{\delta \theta_i} \mathcal{L}^{(k)}[t],$$

where $\theta_i^{(k)}[t]$ is the i -th parameter after the t -th iteration of context k , and $\mathcal{L}^{(k)}[t]$ is the total loss (including the surrogate loss) at the t -th iteration of context k [14], [29]. Similar to EWC, SI also discourages radical changes in important parameters. However, while EWC uses Fisher information to estimate the importance of parameters, SI calculates the importance by accumulating the contribution of each parameter to the loss reduction over the course of training [14].

Learning without Forgetting

Similar to other regularization methods, the loss function of LwF also includes a regularization loss— $\mathcal{L}_{\text{LwF}}(\theta)$ [10]. According to the original paper [10], during context K , $\mathcal{L}_{\text{LwF}}(\theta)$ is defined as knowledge distillation loss [7]:

$$\mathcal{L}_{\text{LwF}}(\theta) = -\lambda \sum_o p_{\hat{\theta}^{(K-1)}}^T(o | x) \log [p_{\theta}^T(o | x)],$$

where λ is the regularization strength (set to 1 based on hyperparameter tuning; see Subsection 4.4 for details), θ is the current parameter, $\hat{\theta}^{(K-1)}$ is the parameter in the end of the last context, and $p_{\theta}^T(o | x)$ is the softmax with temperature, which follows the formula

$$p_{\theta}^T(o | x) = \text{softmax}(z_o^{(x, \theta)} / T) = \frac{\exp [z_o^{(x, \theta)} / T]}{\sum_j \exp [z_j^{(x, \theta)} / T]}.$$

LwF effectively utilizes knowledge distillation to discourage changes in the input-output mapping of the model from the older version of itself [10].

Experience Replay

In ER, a memory buffer with a finite budget is used to store training data from past contexts [21]. During training, the final training dataset of a context is the union of the memory buffer and the dataset of that context. This is different from the algorithm in [20], where a minibatch is sampled from the memory buffer to be combined with the current minibatch. After the training of each context, the memory buffer is updated according to Algorithm 1. ER is a simple and straightforward approach to handle catastrophic forgetting, but it is one of the most effective ones [20], [29].

Algorithm 1 Update Memory Buffer

Input: $\mathcal{D}_{new}$ (new dataset), $\mathcal{M}$ (memory buffer), B (total budget), C (number of contexts so far)

```
new_size  $\leftarrow \lfloor B/C \rfloor$  ; // Compute new size per context
foreach context_index  $\in \mathcal{M}$  do
     $\mathcal{M}[\text{context\_index}] \leftarrow \mathcal{M}[\text{context\_index}][: \text{new\_size}]$  ; // Truncate samples
samples  $\leftarrow \text{RandomSample}(\mathcal{D}_{new}, \text{new\_size})$  ; // Sample subset
new_context_index  $\leftarrow C - 1$  ; // Assign new context ID
 $\mathcal{M}[\text{new\_context\_index}] \leftarrow \text{samples}$  ; // Store samples in memory buffer
```

Averaged Gradient Episodic Memory

Similar to ER, A-GEM also maintains a memory buffer with a finite budget, and it also utilizes Algorithm 1 to manage the buffer [19]. During training, besides the current gradient $g_{current}$ obtained from the loss on the current training batch, A-GEM also computes the replay gradient g_{replay} using the loss on a randomly sampled batch from the memory buffer [19]. The final gradient is computed according to Algorithm 2. If the vector angle between $g_{current}$ and g_{replay} is greater than 90 degrees (i.e., the dot product of $g_{current}$ and g_{replay} is negative), $g_{current}$ is projected to be more aligned with g_{replay} [19]. Otherwise, $g_{current}$ is left unchanged [19]. A-GEM efficiently uses the alignment of $g_{current}$ and g_{replay} to ensure parameters are updated in the correct direction, which effectively mitigates catastrophic forgetting [19].

Algorithm 2 Gradient Calculation and Projection in A-GEM [19]

Input: B_n (current batch), $\mathcal{M}$ (episodic memory), θ (parameters)

```
 $B_M \leftarrow \text{Sample minibatch from } \mathcal{M}$  ; // Sample memory batch
 $\gamma \leftarrow 1 \times 10^{-7}$  ; // Small constant for numerical stability
 $g_{current} \leftarrow \nabla_{\theta} \mathcal{L}(B_n)$  ; // Compute gradient on current batch
 $g_{replay} \leftarrow \nabla_{\theta} \mathcal{L}(B_M)$  ; // Compute reference gradient
if  $\langle g_{current}, g_{replay} \rangle \geq 0$  then
     $g_{final} \leftarrow g_{current}$  ; // No projection needed
else
     $g_{final} \leftarrow g_{current} - \frac{\langle g_{current}, g_{replay} \rangle}{\|g_{replay}\|^2 + \gamma} g_{replay}$  ; // Project gradient
return  $g_{final}$ 
```

4 Experiments and Results

This section covers the results of the experiments. The section is divided into five subsections. Subsection 4.1 provides an overview of the dataset, including the context division method and data processing. Subsection 4.2 describes the setup of the experiments. Subsection 4.3 introduces the evaluation metrics that are used

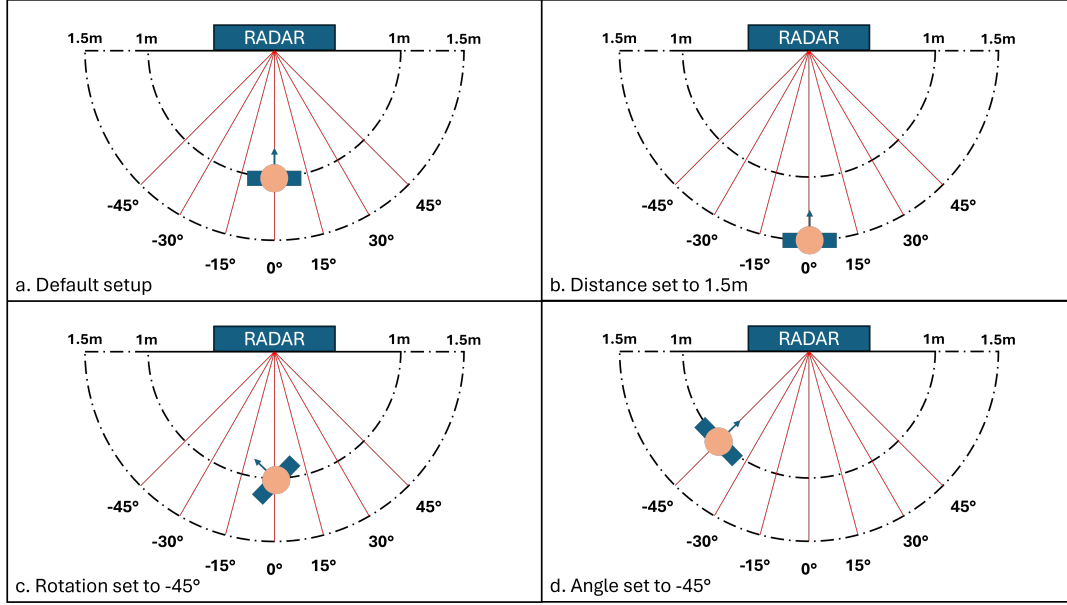


Figure 2: Context setup illustration.

to assess the performance of the methods. Subsection 4.4 demonstrates the setup for hyperparameter tuning. Subsection 4.5 presents and analyzes the results. The interpretation of the results is not included in Subsection 4.5 but in Subsection 5.1 of Section 5.

4.1 Dataset

The dataset used in the experiments consists of motions captured by a radar, which classifies the problem as a depth-based gesture recognition problem. The dataset contains 15 967 samples. After data collection, each sample is processed and converted to a sequence of 28 frames, and each frame includes a point cloud of 120 points. The point cloud is a format that represents a set of points in a three-dimensional space [40]. In the dataset, the sequence of point clouds represents the movement of the participants' fingers. Each point, besides having a three-dimensional coordinate, can also have other attributes (e.g., intensity and color) [40]. In particular, the dataset includes speed and amplitude as additional attributes. Speed indicates the instantaneous speed of the point cloud and amplitude indicates the strength of the reflected radar signal.

The dataset contains 16 gestures (i.e., classes) and is divided into 16 contexts with varying rotation, angle, distance, and set of participants. Rotation refers to the orientation of the participants' body relative to the radar; angle indicates the relative viewing direction from the radar to the participants; distance denotes the space between the radar and the participants; and there are two sets of participants. The default setup uses the first set of participants, and sets the rotation to 0° , the angle to 0° , the distance to 1 meter (see Figure 2a). These values are applied to all contexts unless otherwise specified. From Context 1 to Context 15, only the first 12 classes are presented. The first context is the base context, where the default settings are applied. The second

context also applied the default settings; however, the data is captured from a different set of participants. In the third context, participants are 1.5 meters away from the radar (see Figure 2b). From Context 4 to Context 9, the rotation is set to 15, 30, 45, −10, −30, and −45, respectively (see Figure 2c). From Context 10 to Context 15, the angle is set to 15, 30, 45, −10, −30, and −45, respectively (see Figure 2d). In Context 16, only classes 13 to 16 are present, with the default setup.

4.2 Experiments Setup

The dataset is split into a training dataset with 60% of the data, a validation dataset with 20% of the data, and a test dataset with 20% of the data. In each epoch, the model is trained on the training dataset, and the evaluation metrics are collected with the validation dataset. After all epochs are finished, the model is tested using the test dataset to evaluate its final performance.

The model is trained for 14 epochs for each context. Since there are 16 contexts, the total number of epochs is 224. For the Joint method, where there is only one context with the full dataset, the model is also trained for 224 epochs. In the experiments, cross-entropy loss is used as the base loss function, which is standard for classification problems. For methods that require a memory buffer, the buffer size is set to 300 samples, which is approximately 3.1% of the size of the training dataset.

4.3 Evaluation Metrics

To assess the continual learning capacities of the compared methods, three evaluation metrics are utilized in this thesis: **Final Test Accuracy (FTA)**, **Seen-Context Accuracy (SCA)** across contexts, and **Current-Context Accuracy (CCA)** across contexts. All metrics are based on accuracy, which is formularized as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}.$$

FTA evaluates the model based on its accuracy when tested on the test dataset. SCA captures the performance of the model on the union of the validation datasets of the seen contexts, including the current one. On the other hand, CCA assesses the model using solely the validation dataset of the current context. While FTA provides an overview of the final performance of the model, SCA evaluates the ability of the model to retain learned information from the previous contexts, and CCA reflects the ability of the model to adapt to new training data. Furthermore, for each compared methods (i.e., EWC, SI, LwF, ER, and A-GEM), the differences between the evaluation metrics of the method and of the None method are also calculated, which is referred to as FTA-diff, SCA-diff and CCA-diff. For example, FTA-diff of EWC is computed as

$$\text{FTA-diff}_{\text{EWC}} = \text{FTA}_{\text{EWC}} - \text{FTA}_{\text{None}}.$$

In the experiments, for the Joint method, SCA and CCA evaluations are conducted at the end of the training. For other methods, these metrics are measured after the training of each context.

4.4 Hyperparameter Tuning

For EWC, SI, and LwF, the regularization strength λ is tuned on a logarithmic scale, and the candidate values are 0.01, 0.1, 1, 10, and 100. For each candidate value, the model is trained on 144 epochs (i.e., nine epochs per context). The evaluation metrics used to select the best hyperparameter value are Average CCA and Average SCA, which is the average of CCAs and SCAs across all contexts, respectively. The chosen value for each method is determined by the sum of the two metrics. For the hyperparameter tuning results, see Appendix C.

4.5 Results

In the remainder of this thesis, the term **regularization-based methods** refers to EWC, SI, and LwF; the term **continual learning methods** refers to all methods apart from the Joint method and the None method; the term **incremental methods** refers to all methods except the Joint method; the term **domain-incremental phase** refers to the phase from Context 1 to Context 15; and the term **class-incremental phase** refers to Context 16.

Method	FTA	FTA-diff
Joint	98.8%	
ER	85.0%	+25.3%
A-GEM	81.9%	+22.2%
SI	66.7%	+7.0%
LwF	64.8%	+5.1%
EWC	63.8%	+4.1%
None	59.7%	0.0%

Table 2: Final Test Accuracies of Compared Methods and Their Differences from the None Baseline

Table 2 presents the FTA of the compared methods in descending order. From the table, it can be observed that the FTAs of all incremental methods are between the higher baseline (i.e., the FTA of the Joint method) and the lower baseline (i.e., the FTA of the None method). Among the incremental methods, ER has the highest FTA with 85.0%, followed by A-GEM with 81.9%, SI with 66.7%, LwF with 64.8%, EWC with 63.8%, and None with 59.7%. Notably, there is a significant gap between the FTAs of the regularization-based methods and of the replay-based methods. While the FTA of ER and A-GEM exceeds 80%, the FTA of EWC, SI, and LwF is less than 70%, and the FTA of None is less than 60%. The results of the FTAs highlighted the superiority of replay-based methods over the None method and regularization-based methods.

The upper subplot of Figure 3 shows the CCA evaluations of the compared methods, and the lower subplot presents the difference in CCAs between the None method and other incremental methods. The detailed version of Figure 3 is provided in Appendix A. Apart from being surpassed by two replay-based methods in Context 10, the Joint method outperforms all other methods in all other contexts. Generally,

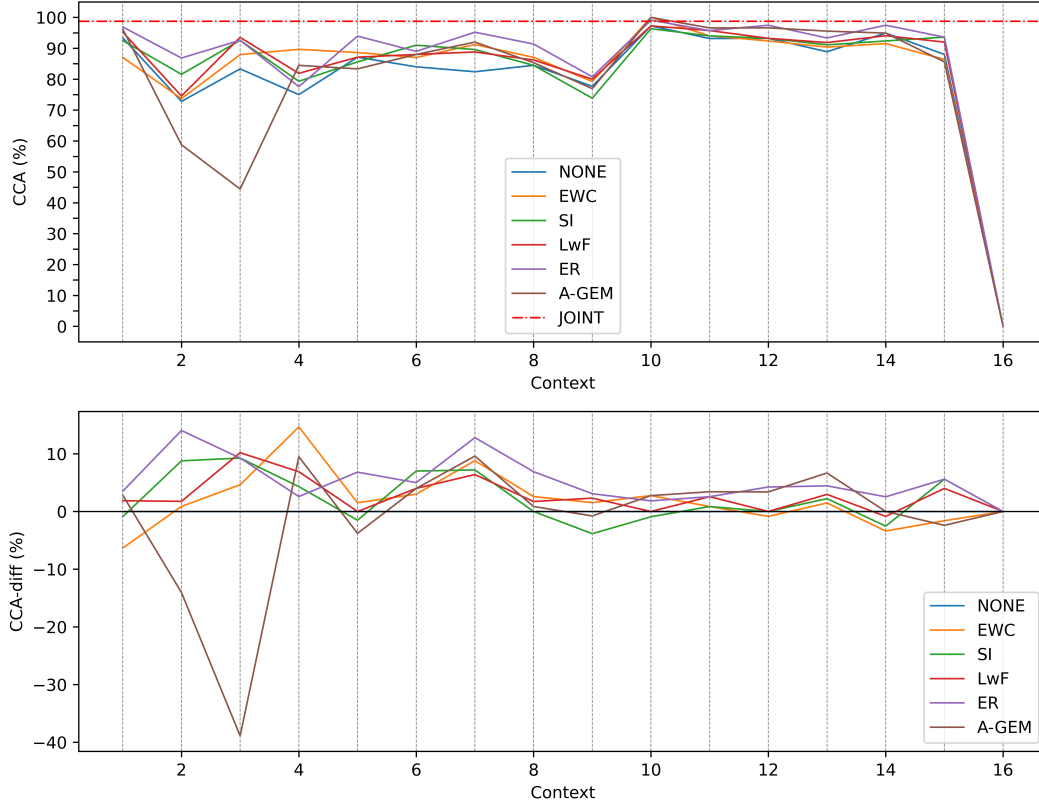


Figure 3: Current-Context Accuracies of Compared Methods in Contexts

most methods manage to maintain a CCA above 70% in the domain-incremental phase, before dropping to 0% in the last context. In particular, ER shows excellent performance by being on the top in multiple contexts and being the only incremental method to match or surpass the None method in all contexts. A-GEM experiences a drop in contexts 2 and 3, then recovers and reaches its peak at Context 10 with a remarkable CCA of 100%. Moreover, regularization-based methods show a relatively stable progression of CCA-diffs across contexts; and they managed to exceed the None method in at least nine contexts. Among these methods, EWC stands out as the method with the best CCA in Context 4. Apart from this case, the CCA-diffs of all regularization-based methods in all contexts fall in the $\pm 10\%$ range. The results of the CCAs highlighted the superior performance of ER and the unremarkable performance of other compared methods.

The upper subplot of Figure 4 illustrates the SCA evaluations of the compared methods, and the lower subplot presents the difference in SCAs between the None method and other incremental methods. The detailed version of Figure 4 is provided in Appendix B. In this figure, no incremental method manages to outperform the Joint method in any context. Meanwhile, ER once again shows excellent performance by surpassing all incremental methods in all contexts. This is followed by A-GEM, which demonstrates a remarkable trajectory of SCA and outperforms all regularization-based methods and the None method from the second to the last context. Furthermore, the

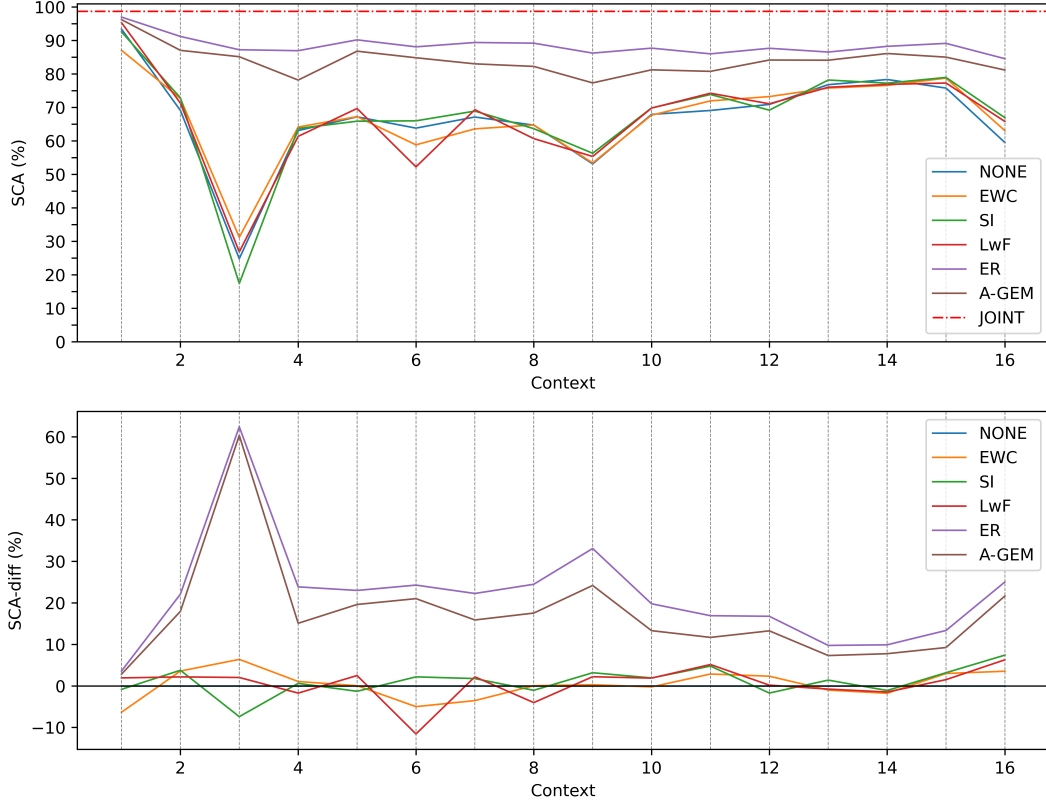


Figure 4: Seen-Context Accuracies of compared methods in contexts.

overall trends in the SCAs of the None method and regularization-based methods aligned closely with each other. They experience a drop in the second context, reach their trough in the third context, then recover and remain in the 50%–70% region in the subsequent contexts. Notably, these behaviors are similar to the behavior of A-GEM in the upper subplot of Figure 3. Furthermore, for most contexts, the SCA-diffs of regularization-based methods stay in the $\pm 10\%$ region, while the SCA-diffs of replay-based methods exceed 10%. The results of the SCAs further demonstrate the excellence of ER and the modest performance of regularization-based methods. In addition, they also highlight the outstanding outcomes achieved by A-GEM.

5 Discussions

This section discusses the results of the experiments. The section is divided into two subsections. Subsection 5.1 provides an interpretation of the result. Subsection 4.2 demonstrates the limitations of this thesis and the directions for future work.

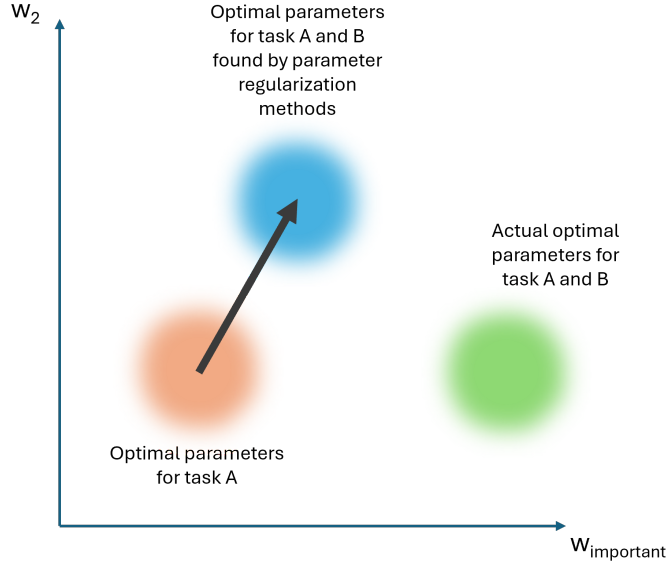


Figure 5: Task-specific parameter regions for parameter regularization methods.

5.1 Interpretation of Results

Apart from being surpassed by ER and A-GEM in Context 10 for the CCA metric, the Joint method, which serves as the higher baseline, outperforms all incremental methods in both SCA and CCA in most contexts. This suggests that while continual learning helps save time and computational resources, retraining with old and new data typically results in a more accurate model.

However, surprisingly, the None method, which serves as the lower baseline, performs better than regularization-based methods in some contexts. For SI and EWC, this suggests that radical changes in important parameters may not have a negative effect on the performance of the network. A parameter may be important for a task, but maintaining its value could be detrimental to the learning of the model in subsequent tasks (see Figure 5). In other words, a set of important parameters having certain values may be optimal for a task, but when considering the joint dataset of all contexts, the optimal values of the same set of parameters might significantly differ. For LwF, being outperformed by the None method in some contexts suggests that using the inputs of the current context to discourage changes in the input-output mapping of the model from the older version of itself is not efficient, especially when there is a significant difference in the data distribution across contexts. Since the old version of the model is not trained on the data of the current context, the input-output mapping of the old version of the model may be uninformative or misleading. Therefore, having it as a soft target to calculate knowledge distillation loss may have a negative effect on the learning ability of the model.

Furthermore, it can be observed that ER and A-GEM outperform all other incremental methods in the metric SCA from the second to the last context and in the metric FTA. This indicates that having access to data from past contexts during the training of the current context is beneficial. For ER, having past data enable the ability

to replay using raw data, which helps the model account for the data distribution of all seen contexts. For A-GEM, thanks to the gradient calculated from the data in the memory buffer, the model can safely update the parameters without harming the performance on past tasks.

Notably, in Context 3, regularization-based methods and the None method experience a significant drop in SCA, while the CCA of A-GEM plummets. The only method that performs well in the third context is ER. Context 3 is the context in which participants stand 1.5 meters away from the radar, which might be due to sparse data captured by the radar. Due to sparse data, the data distribution of the third context might be substantially different from that of the first two contexts, resulting in difficulties of the network in remembering information from the first two contexts during the third one. This explains the drop in the SCA of regularization-based methods and the None method, as they manage to adapt to the data of Context 3 but at the expense of past information. On the other hand, A-GEM manages to maintain an SCA above 80% but with a significant decrease in CCA. This demonstrates that in Context 3, ER finds the balance in the stability-plasticity dilemma, while A-GEM trades plasticity for stability and regularization-based methods trade stability for plasticity.

Another notable phenomenon is the drop to 0% in the CCAs of all incremental methods in the last context, where Classes 13 to 16 are introduced. A theory for this is that after being trained on datasets with 12 classes for more than 200 epochs, the model has learned to assign low values to the logits corresponding to the last four classes. Hence, the model is heavily biased toward predicting samples belonging to the first 12 classes.

5.2 Limitations and Directions for Future Research

In this thesis, experiments have been conducted to benchmark the performance of different continual learning approaches in RF-based gesture recognition. Due to time and resource constraints, there are some limitations that need to be acknowledged. First, only five continual learning methods are included in this work. Future work could explore other existing methods (e.g., Incremental Classifier and Representation Learning [12] and Functional Regularization of Memorable Past [26]) or develop new methods. Furthermore, all three evaluation metrics in this thesis are limited to accuracy. While they provide an overview of the performance of the methods, there are other metrics that can provide a more comprehensive insight into the behaviors of the compared methods. Future work could consider including other evaluation metrics, such as Forgetting [15], Intransigence [15], Backward Transfer [16], and Forward Transfer [16]. In addition, this thesis failed to compare the training time of the compared methods since the experiments are run on different devices with varied computation power to maximize time efficiency. Future work could standardize the hardware to enable a comparison of the computational efficiency of each method.

6 Conclusion

In conclusion, this thesis compares the performance of different continual learning methods when applied to RF-based gesture recognition. The compared continual learning methods include three regularization-based methods and two replay-based methods. Throughout the experiments, it can be observed that the two replay-based methods performed better than the three regularization-based methods. This suggests that retaining past data (i.e., maintaining a memory buffer) plays a significant role in continual learning, while having a regularization term may be insufficient to mitigate catastrophic forgetting. Surprisingly, the performance of all continual learning methods drops significantly in the last context, where new classes are introduced. This is because the model is heavily biased toward predicting samples belonging to the first 12 classes. However, this thesis has some limitations regarding the number of compared methods, the range of evaluation metrics utilized, and the benchmarking of computational efficiency due to time and resource constraints. Future work should explore more methods or develop new ones, employ more evaluation metrics, and standardize hardware to allow the measurement of training time.

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A Detailed Current-Context Accuracies (CCAs) Results Figure

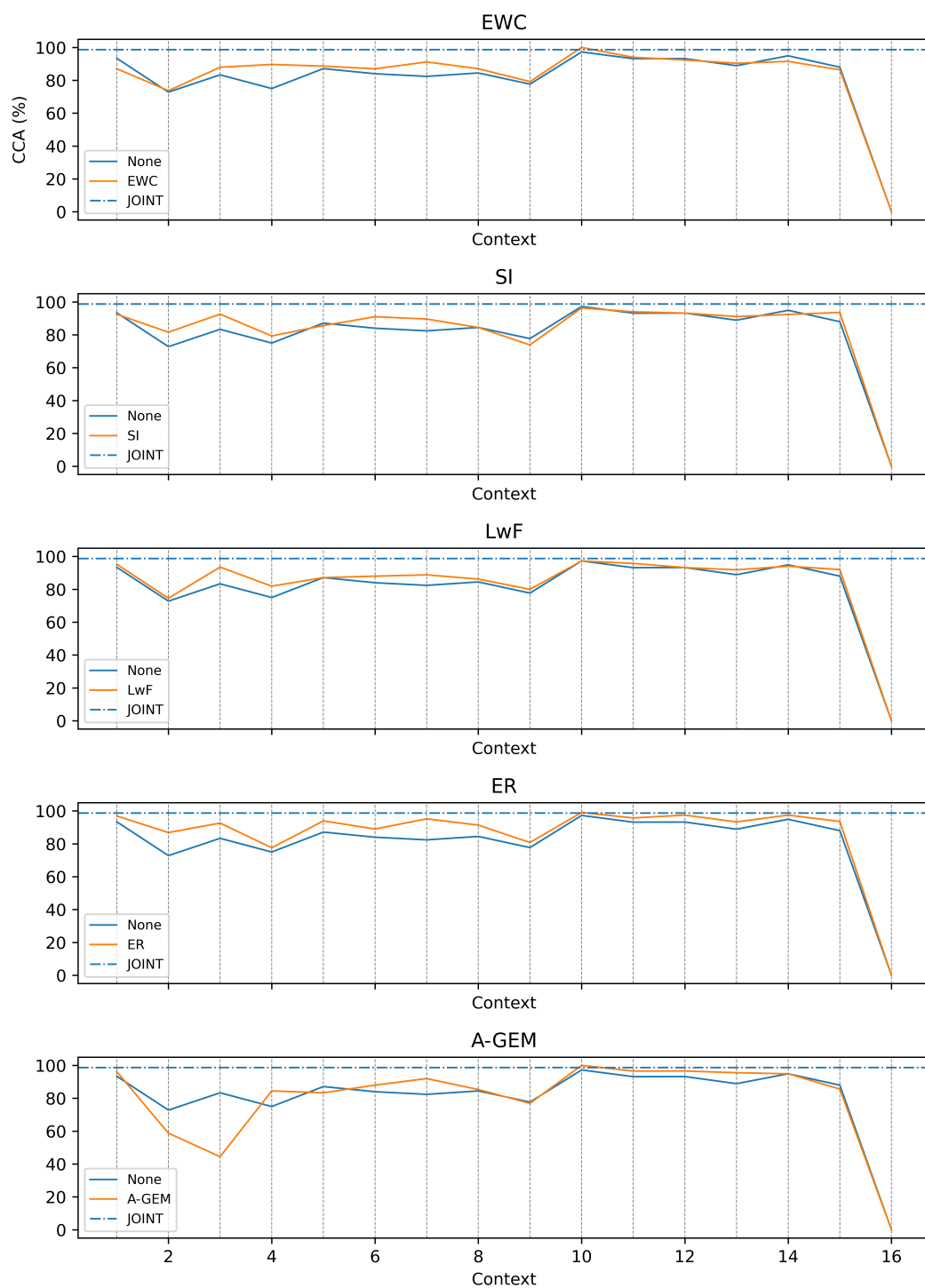


Figure 6: Detailed Current-Context Accuracies (CCAs) Results Figure

B Detailed Seen-Context Accuracies (SCAs) Results Figure

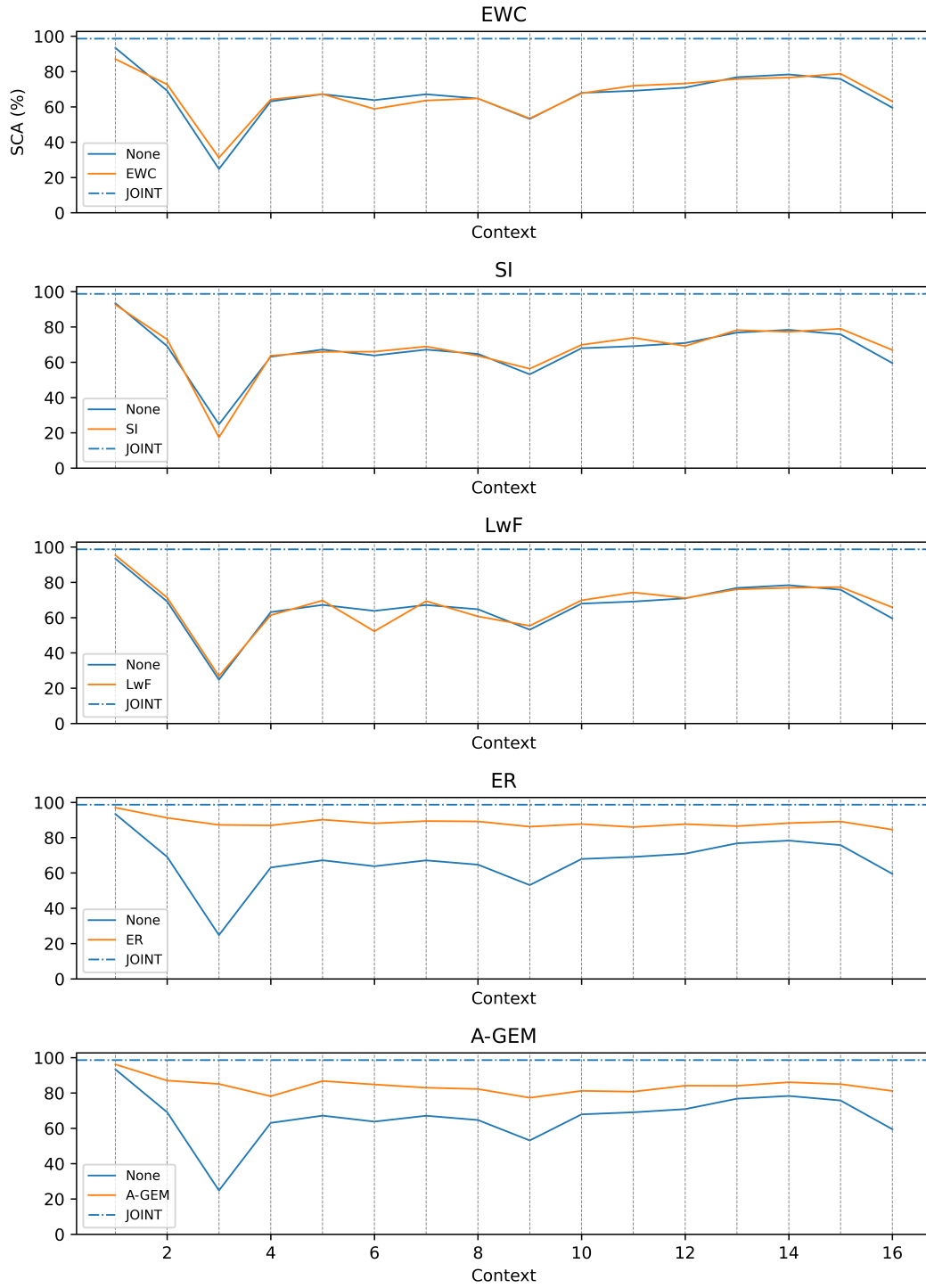


Figure 7: Detailed Seen-Context Accuracies (SCAs) Results Figure

C Hyperparameter Tuning Results

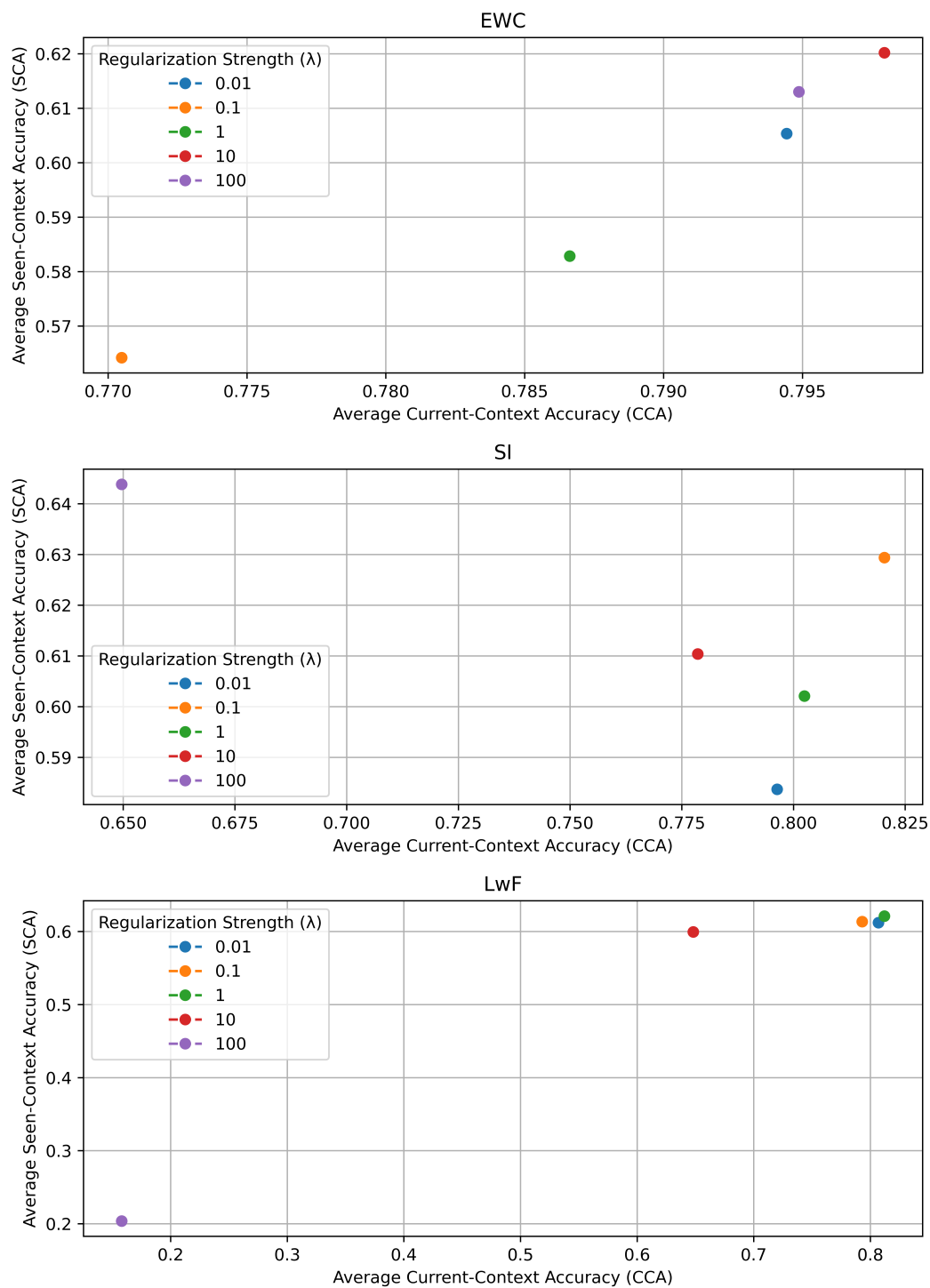


Figure 8: Hyperparameter Tuning Results